Annapureddy Sai Ramyasri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.58
Bhashyam Junior College Guntur	Class XII (Board of Intermediate Education Andhra Pradesh)	2024	98.10%
Bhashyam High School	Class X (Board of Secondary Education Andhra Pradesh)	2022	98.67%

PROJECTS

Solar Cell & Diode I-V Characterization and Automatic Solar Tracking | Arduino, Arduino IDE, Python

- Built an Arduino-based measurement setup to obtain the voltage-current characteristics of solar cells and semiconductor diodes by varying the applied input using PWM.
- Collected voltage and current measurements from the experimental setup and processed the recorded data in Python to generate I-V curves and analyze device characteristics.
- Calculated and monitored output power from the measured voltage and current values, providing real-time observations of the solar panel's electrical performance.
- Developed an automatic solar tracking mechanism using LDR sensors and a servo motor to continuously adjust panel orientation toward the incident light and evaluate the improvement in power generation.

Analog Lock-in Amplifier Design and PCB Implementation | Altium Designer, KiCad, LTspice

- Designed and simulated an analog lock-in amplifier for extracting weak signals from noisy environments using synchronous detection.
- Implemented the signal chain comprising an **AD633 analog multiplier**, **ADA4610 operational amplifier**, and **Sallen-Key low-pass filter** for signal recovery.
- Designed and fabricated the PCB in **Altium Designer**, incorporating component placement, routing, grounding, and power distribution for low-noise analog operation.
- Verified the circuit functionality through **KiCad** and **LTspice** simulations under noisy input conditions prior to PCB fabrication.

Regression Algorithms from Scratch | Python, NumPy, Matplotlib

- Implemented Linear Regression and Logistic Regression from scratch using Python and NumPy without ML libraries.
- Developed Gradient Descent to optimize model parameters and minimize loss functions.
- Implemented Mean Squared Error (MSE) for Linear Regression and Binary Cross-Entropy (BCE) for Logistic Regression.
- Visualized model predictions and regression results using Matplotlib.

LearnFlow - Learning Resource Management Web Application | React, TypeScript, FastAPI, SQLite, SQLAlchemy, JWT

- Built a full-stack web application for organizing YouTube videos, PDFs, and coding problems into custom learning collections and tracking progress across different resources.
- Developed REST APIs using FastAPI and SQLAlchemy for user accounts, collections, and learning resources, with JWT-based authentication and protected endpoints for user-specific data.
- Designed the SQLite database layer with relational models for users, collections, and resources, and implemented CRUD operations for creating, editing, deleting, and organizing learning content.
- Built a React and TypeScript dashboard with confidence ratings and revision reminders to help users identify weaker resources and prioritize what to revisit.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures & Applications, Machine Learning

Mathematics: Calculus-I, Calculus-II, Differential Equations, Probability and Random Variables, Complex Variables, Introduction to statistics.

Electronics: Semiconductor Device Fundamentals, Analog Electronics, Circuits and Network Analysis, Digital Systems, Signals and Systems, PCB Design and Layout for Electronic Systems, Semiconductor Devices Lab, Introduction to CMOS Processing

SKILLS

Programming Languages - Python, C, C++, JavaScript, TypeScript, SQL

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS)

Developer Tools: Git, GitHub, Linux, VS Code, PyCharm

Databases: SQLite, MySQL

Frameworks & Libraries: React, FastAPI, SQLAlchemy, OpenCV, MediaPipe, NumPy

Web Technologies: HTML , CSS

POSITIONS OF RESPONSIBILITY

Elektronica (Electronics and Signals Club) Core Member - Assisted in conducting a technical session for first-year students and organizing the Freshers' Orientation program.

Inventure Club (Yoga and meditation Club) Core Member - Assisted in organizing weekly meditation sessions and managed social media promotions for club activities

EXTRACURRICULAR

NSS Volunteer

Volunteered in Medical Support Team in IITH Half Marathon Event Dec 2024